

Huawei's "Tau" Scaling Law - Not Yet a DeepSeek Moment for China Semi

Huawei's top-down system topology aims at making chips without EUV that could perform like 1.4nm chips (2031). One key is Logic Folding, which connects two chips using hybrid bonding as a routing resource. This could reduce die size (potentially higher yield) and shorten transmission distance (higher performance). But we see challenges in heat dissipation, wafer variations, hybrid bonding density and more. Positive for EDA, WFE, foundries and optics.

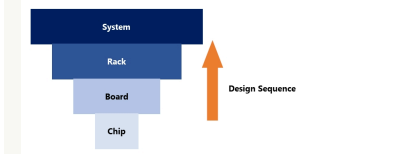
Huawei's (HW) τ Scaling Law reframes chip performance from node-based scaling to system-level time optimization. Presented at ISCAS 2026, Huawei's τ / Tau Scaling Law shifts focus from traditional geometry scaling / Moore's Law toward a holistic, top-down design methodology. Instead of treating process node size, such as "x nm," as the primary proxy for performance, HW argues that time-to-computation / τ should become the key metric. Performance gains can come not only from transistor shrink, but also from circuit design, chip architecture, packaging, interconnect, and system-level optimization. This roadmap is especially relevant as Moore's Law slows and Huawei faces restricted access to leading-edge process technologies.

Logic Folding is HW's most notable example of using 3D design to reduce critical-path delay. In chip design, performance is capped by the slowest signal path, known as the critical path. Traditionally, engineers shorten this path through better place-and-route or by moving to a more advanced process node. HW's Logic Folding concept attempts to "fold" a 2D chip layout into a 3D structure by stacking logic layers, allowing connected elements to be placed closer together vertically. Although this resembles existing 3D-IC and chiplet approaches, HW's vision is more integrated: instead of treating chiplets as separate black boxes connected through limited I/O, Logic Folding aims to co-optimize multiple layers as one unified design.

Full Logic Folding could create major opportunities, but it faces significant technical challenges. The roadmap can be viewed in three stages: basic 3D stacking (ie, 3D-IC proposed by TSMC/Intel), co-simulation-level Logic Folding, and ultimately co-design-level Logic Folding. The most advanced stage would require EDA tools to treat hybrid-bonding layers as routing resources, effectively moving chip design from 2D into true 3D. This would require a major redesign of the EDA toolchain and create demand for 3D-aware place-and-route, timing, thermal, and reliability tools. However, major obstacles remain, including overheating, wafer-to-wafer variation, limited hybrid-bonding density, TSV reliability, yield risk, and manufacturability. As a result, full commercial viability may take a long time.

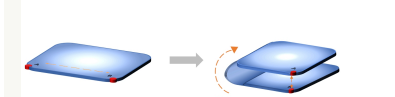
We examine how this roadmap would impact different supply chain players. HW also extends τ scaling beyond chips through its System-as-One-Chip concept, using Unified Bus and Hi-ONE optical interconnects to optimize the entire AI system. This could simplify data-center architecture, reduce latency, improve heat dissipation, and lower total cost of ownership by replacing fragmented protocols with a more unified topology. If successful, the roadmap could benefit EDA vendors (higher demand but needs innovations), hybrid bonding equipment suppliers, optical interconnect players, and advanced node foundries. However, it could pressure AI server ODMs (simplified configuration), cooling (reduced demand) and power suppliers (lower power density), and independent Chinese AI chip design houses, if HW gains a proprietary system-level performance advantage. This roadmap will likely have global impact, as non-Chinese players could also pursue this approach and geometry scaling simultaneously.

Chart 1 - Bottom-up System Design



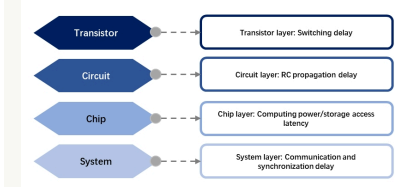
Source: Jefferies

Chart 2 - Space Folding



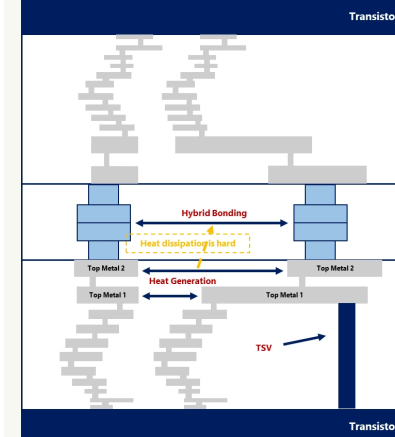
Source: Jefferies

Chart 3 - Core Concept of τ (Tau) Scaling



Source: Jefferies

Chart 4 - Face to Face Hybrid Bonding



Source: Jefferies

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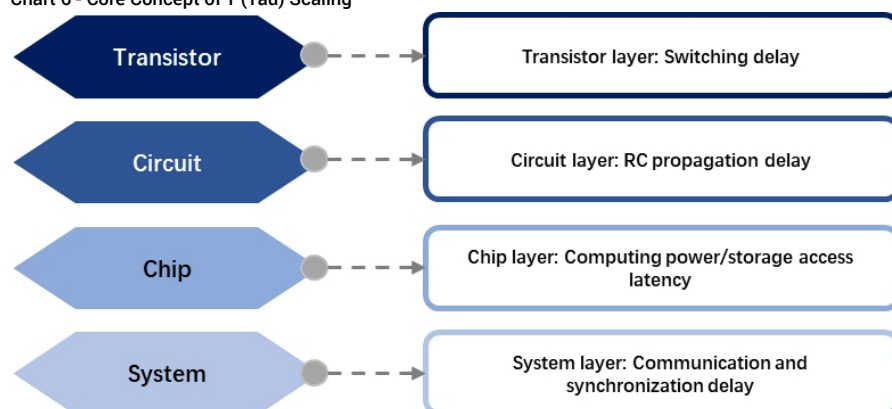
On 2026 May 25, Huawei presented “**τ (Tau) Scaling Law**” at the IEEE International Symposium on Circuits and Systems (ISCAS) 2026. The company turned their design strategy to a **more holistic/top-down approach** from a bottom up/lego-like approach, which the industry has been using for decades. The change/innovation is more on design methodology rather than manufacturing technologies, so it offers the industry a way to pursue better performance without accessing the latest process node tech. Although there are still technological challenges to overcome, the proposal already made a “dimensional” impact on computing, including AI, system design.

Chart 5 - τ (Tau) Scaling

$$\tau = f(\tau_{transistor}, \tau_{Circuit}, \tau_{Chip}, \tau_{System})$$

Source: Jefferies

Chart 6 - Core Concept of τ (Tau) Scaling



Source: Jefferies

Huawei first proposed “**τ scaling**” as the new metric to replace traditional “**geometry scaling/Moore’s law**” to evaluate a new system. It is because while geometry scaling can lead to computing performance gain, it is not the only factor. Device shrink (geometry scaling), circuit design, chip design and system design improvement can all improve the performance of a compute system. From Huawei’s perspective, as Moore’s law slows down, geometry shrink will create less incremental performance gain, so geometry scaling (such as x nm) can no longer be used as the only proxy of performance. They believe time, which shows how fast a computation can be done, should be the new metric of evaluating systems performance.

This concept is not entirely new. Due to geopolitical tension, Huawei has been trying to use better system design to overcome US restrictions that constrain their chip performance. However, in IC design, chip performance is usually balanced against other requirements, such as power efficiency, heat control, cost, reliability, and many more. We believe Huawei, as one of the best chip design houses globally, clearly understands that. Thus, we see the announcement of this “τ Scaling” roadmap a declaration that **Huawei would pursue higher chip performance at all costs.**

In the presentation, Huawei used some examples to demonstrate how this new roadmap will be executed. Among all the examples, “**Logic-folding**” is the most eye-catching one.

Sci-Fi fans must be familiar with “space folding” and “wormhole”. A typical demonstration for space folding is to draw two points on a paper. On a 2-dimensional paper, the closest distance of the two points is the straight line between the two. However, in a 3-dimensional space, the paper can be folded so the two points overlap each other and the distance between them is practically zero.

When moving from a lower dimension world to a higher one, space can be folded, and distance can be shortened.

Chart 7 - Space Folding



Source: Jefferies

Chip-performance is capped by the slowest part of the chip, which is determined by the distance between two connected transistors (because signal's travel time will be longer for longer distance). The industry's term for the longest connection is "critical path". In other words, if engineers can shorten the "critical path", they can boost the performance of the chip. Traditionally, this can be achieved by:

1. **Optimizing Place & Route (P&R):** It means cleverly placing all related transistors closer to each other to minimize critical path. This is one of the most critical functions of EDA tools.
2. **Using more advanced node** manufacturing tech: transistor size and wire lengths among them naturally shrink as the chip migrates to more advanced node. In consequence, critical path also naturally shortens.

Both methods aim to shorten critical paths in a two-dimensional (2D) space because modern "chip" design is essentially a 2D design, where all transistors are placed on the same surface. If engineers have already done their best in P&R, traditionally, the only way to improve further is to adopt more advanced node. Therefore, the industry sees geometry scaling the proxy and metric of chip performance improvement.

With the holistic approach, Huawei argues that design engineers should only focus on the "end performance (τ)" instead of geometry scaling, which is just one of the factors that might lead to better performance. They transformed the 2D space in a chip into a three-dimensional (3D) space by stacking one chip on top of another. Just like folded paper we discussed above, the space is folded, and the longest distance (critical path) is shortened. Hence the performance (τ) increases without any change in geometry scale.

For many tech investors, this may sound very similar to the 3D-IC concept that TSMC and Intel proposed long time ago. However, there are some significant differences.

An analogy for the differences is connecting two apartment units with one directly above the other to create a duplex. If you ask two interior design teams to work on each unit separately, there would be some redundancies and suboptimal design between the two units. If a single team takes a holistic view to design the entire duplex, the team will likely be able to utilize the entire space more efficiently by, for example, putting all bedrooms upstairs, and creating a bigger living room and also a bigger kitchen downstairs.

TSMC's current chip-let/3D-IC approach is similar to the first approach. Each chip-let is self-optimized within its own and use very limited I/O ports to communicate with other chip-lets. In other words, TSMC's job is only to build a staircase to connect the two floors. Each floor is designed by separate teams with possibly different objectives. However, Huawei wants to design the entire duplex with a unified approach and style

We see three levels of 3D chip design:

1. 3D Stacking (Current TSMC's SoIC solutions)

- Each chip-let is a black box. Design engineers only know the I/O information about the chip. The benefit of this is to allow multiple companies' IPs to co-exist in the same 3D chip.
- This is the least integrated solution.

2. Co-simulation-level Logic Folding*

- The box needs to share its model (gate level design information) so that the other chip may use this information to identify critical path.
- EDA tools still do the simulation separately in 2D chips and then optimize by identifying and shortening critical paths via vertical connections.
- This approach is less likely to allow collaboration as no design houses want to share their models (gate level design information), which are considered commercial secrets.
- The existing EDA tools already can achieve this. TSMC already provides PDK and EDA vendors use TSMC's PDK to simulate TSV and related circuit. According to Huawei's paper, we believe they are at this stage.

3. Co-design-level Logic Folding*

- The EDA tools need to consider everything with both floors and units in mind, meaning that EDA tools will need to move from 2D space to 3D space. Conceptually, design engineers will use the hybrid bonding layer as another layer of metal/via. The whole EDA design chain needs to be revamped.
- This is the most integrated solution, and this should be the ultimate goal for logic folding.

*We coined these two terms for convenience of explanation.

Over the past few days, we have discussed Huawei's approach with several industry experts, and learned the following key challenges that Huawei could face in achieving the co-design-level logic folding:

1. Overheating

- Overheating is one of the most obvious issues as the high layer count would trap heat in the core. During the Huawei presentation, it presents a face-to-face structure, which means they use hybrid bonding to fuse the two-face side of wafer, which are the hottest. In the worst-case scenario, this might cause "thermal runaway", where overheating causes higher power consumption, and then higher power consumption causes even higher heat chain reaction until the whole system breaks down.
- Anslys platform (Synopsys) and Celsius platform (Cadence) (EDA tools) will see high demand if logic folding becomes mainstream.

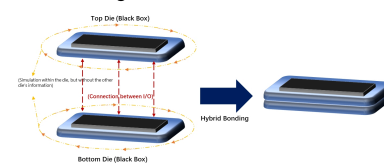
2. Process Control

- Process control can be a serious challenge. As Chinese fabs do not have access to the most advanced equipment, the variation among the wafers (Wafer-to-Wafer variation) can be bigger than normal. However, wafer-to-wafer consistency is critical in co-design-level logic folding, as the two layers (or even more) will be coming from different wafers. Signal transmission timing could be adversely impacted by such variations.

3. Hybrid Bonding Density

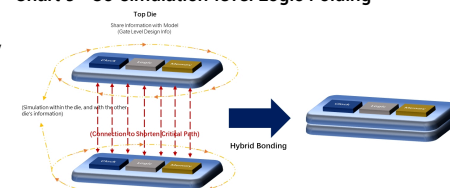
- In the current TSMC chiplet solution, all chips are connected only through their I/Os. Thus, the need for hybrid bonding connection is limited. However, if hybrid bonding

Chart 8 - 3D Stacking



Source: Jefferies

Chart 9 - Co-simulation-level Logic Folding*



Source: Jefferies

Chart 10 - Co-design-level Logic Folding*

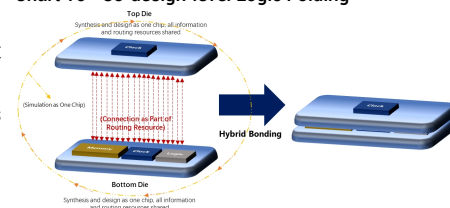
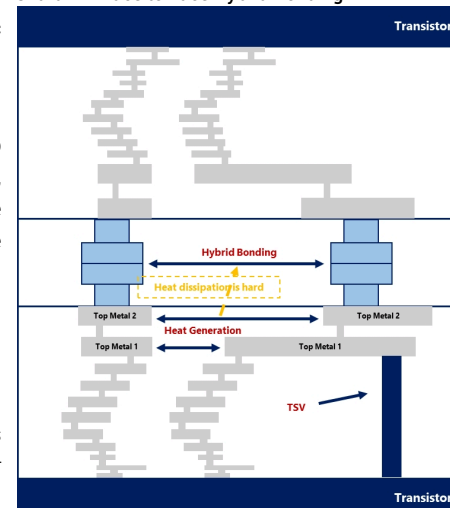


Chart 11 - Face to Face Hybrid Bonding



Source: Jefferies

connections are used as part of routing resources (in the co-design-level), the number and density of hybrid bonding may need to increase exponentially.

- b. Our understanding is that in hybrid bonding layer today, only less than 15% area can be used to do connection. Moreover, the pitch of these copper connections is as big as 5~10um (Mass Production with large chip size). This will make co-design-level logic folding very challenging to achieve.

4. TSV Thickness and Length

- a. During the presentation, Huawei highlighted TSV as a potential roadblock during routing, and they hope to make TSV thinner to liberate routing resources at top metal layers. However, our understanding is that a thin TSV would cause serious reliability and longevity issues.

While most of these challenges already exist in TSMC's 3D Stacking technology, they could become more serious in Logic Folding.

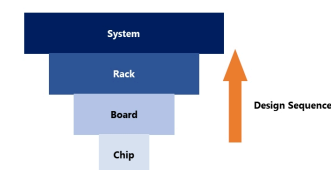
In conclusion, Logic-folding is a great example for Huawei's τ centric holistic design methodology. Instead of struggling to compete in geometry scaling, they might be able to use logic-folding to achieve a better (smaller) τ . However, this is a totally new design concept, which would require new design chain. **Innovations in EDA tools will be required and demand for many more of such tools will be highly likely.** On the other hand, multiple existing challenges in the current 3D stacking technology will likely expand under logic folding. Therefore, the ultimate solution may take a long time to become technologically practical and commercially viable.

Another example of using the τ centric holistic way to make ultimate system design is what Huawei calls System-as-One-Chip, basically using the same holistic approach to design the system, not just the chips. Unified Bus & Hi-ONE (High-density Optical-interconnect-Node Engine) are the key technologies they demonstrate.

Due to the vertically disintegrated semiconductor industry, chip products tend to be built from bottom up and using a multi-layer approach. IC design houses could make the best, most integrated, high-performance chip, and then deliver to board engineers. Board engineers would do the same and then pass on to rack engineers and then the system engineers. Every layer of this supply chain is doing their best to provide the most integrated solutions. The result is an integrated, but not unified, super hot rack solution with all different protocols transmitting data through the layers, thus creating latency.

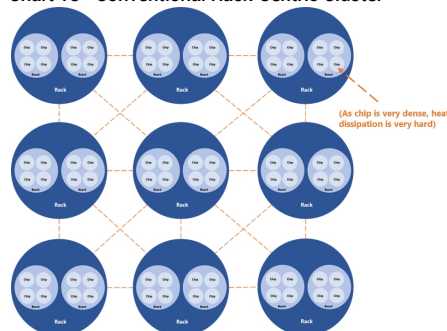
Huawei is using a top-down/holistic approach. It starts with system-level topology, and replaces all different protocols with one single "Unified Bus" ports and then connects the data to in-package-optical-module (Hi-ONE) that allows fast transmission for a distance of 100m. With the new design, we think each node, which is the AI chip, will be far from each other and thus much easier for heat dissipation. In this case, we believe, the data center infrastructure will be much simpler, have lower TCO, and have the best performance possible. Huawei's design focus on the final τ , not the τ in any sub-layers.

Chart 12 - Bottom-up System Design



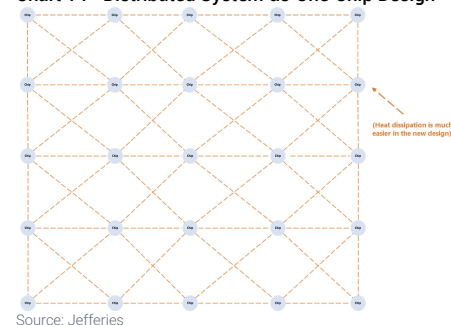
Source: Jefferies

Chart 13 - Conventional Rack-Centric Cluster



Source: Jefferies

Chart 14 - Distributed System-as-One-Chip Design



Source: Jefferies

This is a design methodology that might have global impact. We believe this proposal may not only impact how chips will be designed in China, but could also be a lesson for players outside China to observe and learn what works and what does not. In our view, nothing can stop international supply chain and IC design houses from adopting the same approach. We believe this roadmap will positively impact:

1. EDA Players: EDA vendors need to be ready for 3D, multi-layer P&R capabilities.
2. 3D-IC SPE: Hybrid bonding, CMP, and TSV names are obvious players that might benefit.
3. Fabless Chip Designers (non-China), as they will create more value with the upgrade.
4. Heat-dissipation names: 3D structure is hot. And they need support to dissipate heat. Our understanding is that graphite and SiC substrate could be used for fast heat dissipation.
5. Optical Players: Huawei's Hi-ONE will be key system design innovation., which we believe will lead to the eventual adoption of CPO, enabling smaller chips, higher yield, lower heat and higher transmission speed. This is similar to TSMC's OIO technology.
6. Foundries that have advanced node capabilities such as SMIC and Huali: more powerful chips would boost demand given AI demand, and advanced packaging would create incremental revenue potential

But it could potentially impact the following areas negatively:

1. Server ODMs/AI server component suppliers/liquid cooling players/power supply players: As the new distributed system design reduces rack's and board's power intensity, demand for cooling/power supply/complex AI racks could be reduced longer term.
2. Chinese fabless AI chip design houses: In China, we believe Huawei might lead the technology and stay exclusive at least for a while. This will create a significant advantage for Huawei. Other AI chips players might find it hard to compete.

Company Valuation/Risks

Intel Corporation

Price Target: \$120 implies 58x our CY28E EPS of \$2.06.

Upside:

- Slower-than-expected share loss in datacenter.
- PC TAM higher on Enterprise refreshes and earlier-than-expected AI PC applications.
- Return to capital return commitment with dividend increase.
- Design wins and share gains in datacenter for parallel processing and accelerator products.
- AMD fumbles, leading to share gains for INTC.

Risks:

- Larger-than-expected share loss to AMD and ARM CPUs in datacenter.
- Decline in PC unit TAM and/or ARM disruption.
- Delays to upcoming 18A product launches and/or Foundry customer traction.
- Poor execution of manufacturing process technology transitions.
- Dividend cut further.

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(Article 3(1)e and Article 7 of MAR)

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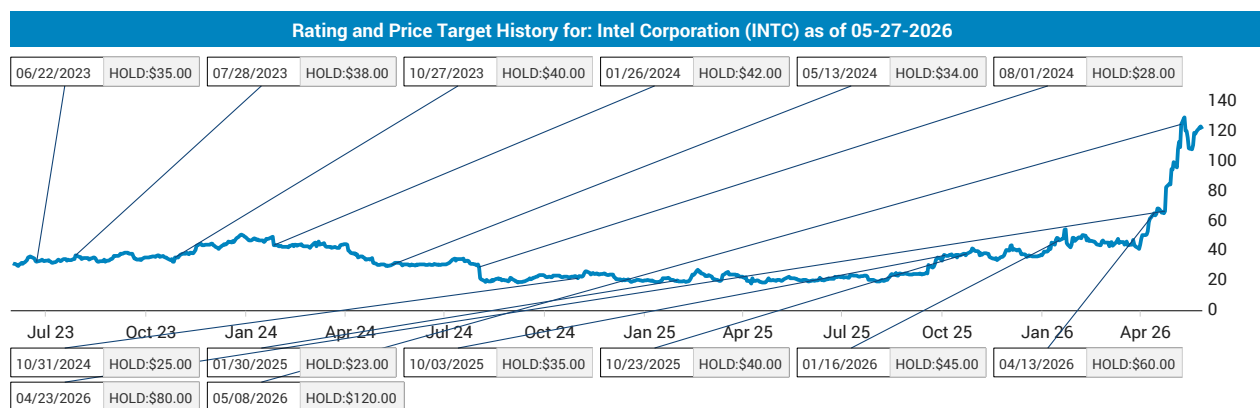
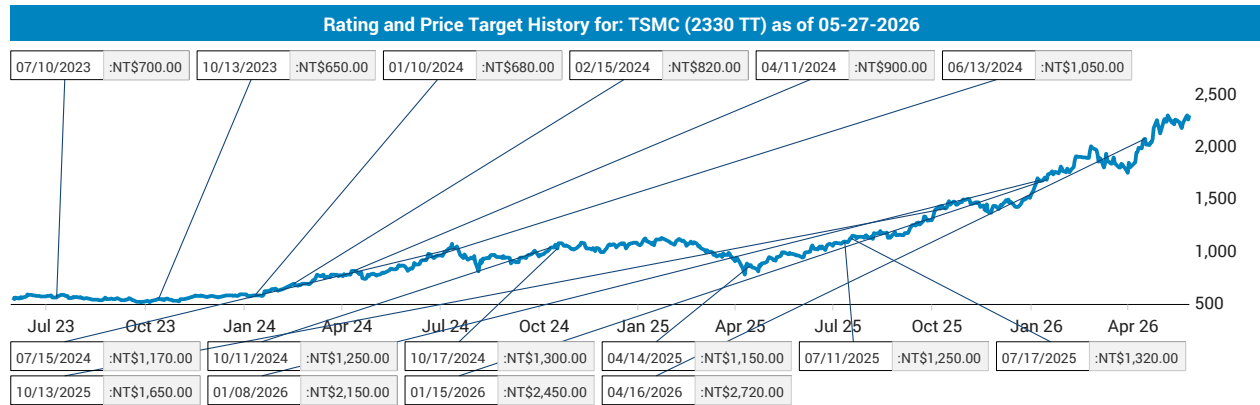
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D: Dropped Coverage

B: Buy

H: Hold

UP: Underperform

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